

Machine Learning Final Project Report

Scholarship Prediction System

Course: Introduction to Machine Learning

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Abstract

This project develops a scholarship prediction system using a supervised machine learning workflow. Starting from the midterm baseline, we rebuilt the pipeline on the v2 dataset and expanded it into a more rigorous final solution. The workflow includes exploratory data analysis (EDA), interaction-based feature engineering, leakage-free preprocessing, model training, systematic comparison, group-based error analysis, model interpretation, and stability validation across repeated splits. We trained four classical models: Logistic Regression, Support Vector Machine (SVM), Random Forest, and XGBoost. On the development set, Logistic Regression and SVM achieved the strongest classification performance, with Logistic Regression providing the best overall balance of predictive quality, stability, and interpretability. The final recommendation is Logistic Regression because it combines the highest ROC-AUC (0.9983) and PR-AUC (0.9977) with a strong F1-score (0.8889) and transparent decision logic suitable for educational policy contexts.

1. Introduction

The goal of this project is to predict whether a student receives a scholarship based on academic and socio-economic attributes. This task is a binary classification problem and is representative of a practical educational decision-support setting where both predictive accuracy and recall are critical. Missing a deserving student (false negative) carries higher institutional cost than a minor false alarm, making model behavior under precision-recall trade-offs particularly important.

The objective of the final project is not only to obtain strong predictive performance, but also to justify modeling choices, compare several algorithms in a principled way, and explain model behavior transparently. Compared with the midterm version, the final project extends the baseline system in several meaningful ways: we use the updated v2 dataset, perform more systematic exploratory analysis, add interaction-based feature engineering, train and compare four models instead of two, report ranking metrics (ROC-AUC and PR-AUC), analyze errors by demographic and academic groups, interpret model outputs, and evaluate stability across multiple random splits.

2. Dataset Description

The dataset contains synthetic student records designed for educational machine learning tasks. It is split into a training set ($n=250$) and a development set ($n=100$).

Feature	Type	Range	Description
id	Integer	1–250	Unique student identifier
gpa	Float	2.00–3.99	Grade Point Average
attendance_rate	Float	0.60–1.00	Proportion of classes attended
study_hours_per_week	Integer	5–39	Self-study time per week
exam_score	Integer	45–99	Final examination score
household_income	Integer	205–2497	Annual family income (USD)
part_time_job_hours	Integer	0–24	Weekly part-time work hours
label	Binary	{0, 1}	Target: 1 = scholarship received

Target Distribution:

The training set exhibits moderate class imbalance: 150 negative samples (60%) and 100 positive samples (40%). The development set follows the same distribution. This imbalance justifies the use of F1-score, PR-AUC, and recall alongside accuracy.

Data Quality:

No missing values were detected across all features. All predictors are numeric, eliminating the need for categorical encoding. Feature scales vary significantly (e.g., gpa vs. household_income), which motivates standardization for scale-sensitive models.

3. Exploratory Data Analysis

We conducted EDA to understand dataset structure, feature ranges, and relationships with the target variable.

3.1 Descriptive Statistics & Scale Differences

Summary statistics confirmed that the dataset is clean and fully numeric. Notably, household_income spans three orders of magnitude larger than other features, while attendance_rate is bounded between 0.60 and 1.00. These scale differences support the use of StandardScaler for linear and distance-based models.

3.2 Target & Feature Distributions

Students receiving scholarships (label=1) are heavily clustered at higher GPA values and higher exam scores. Attendance rate shows a moderate positive association with the target, while part-time job hours exhibit a slight negative trend, likely reflecting time-constraint effects on academic performance.

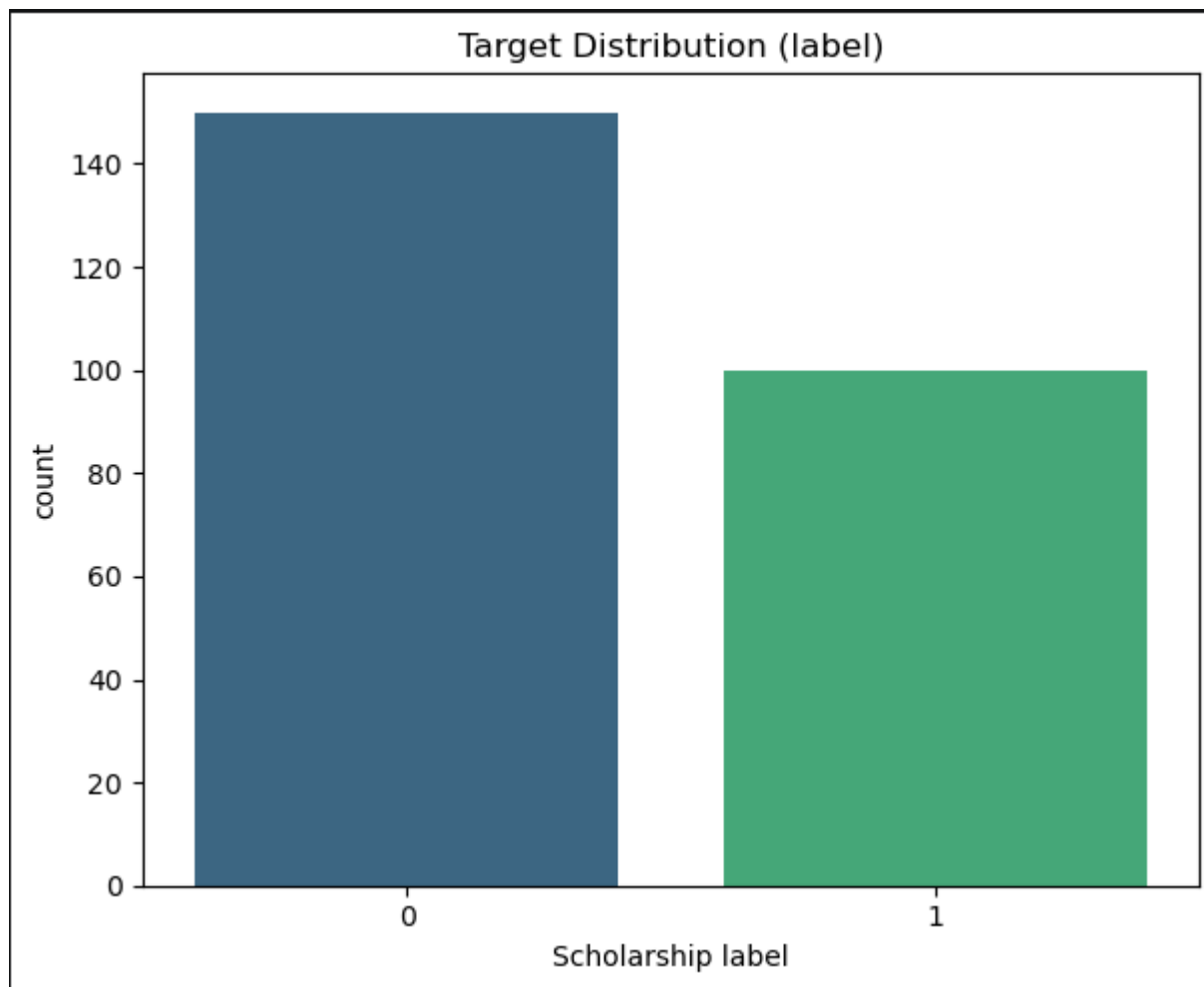


Figure 1. Target label distribution

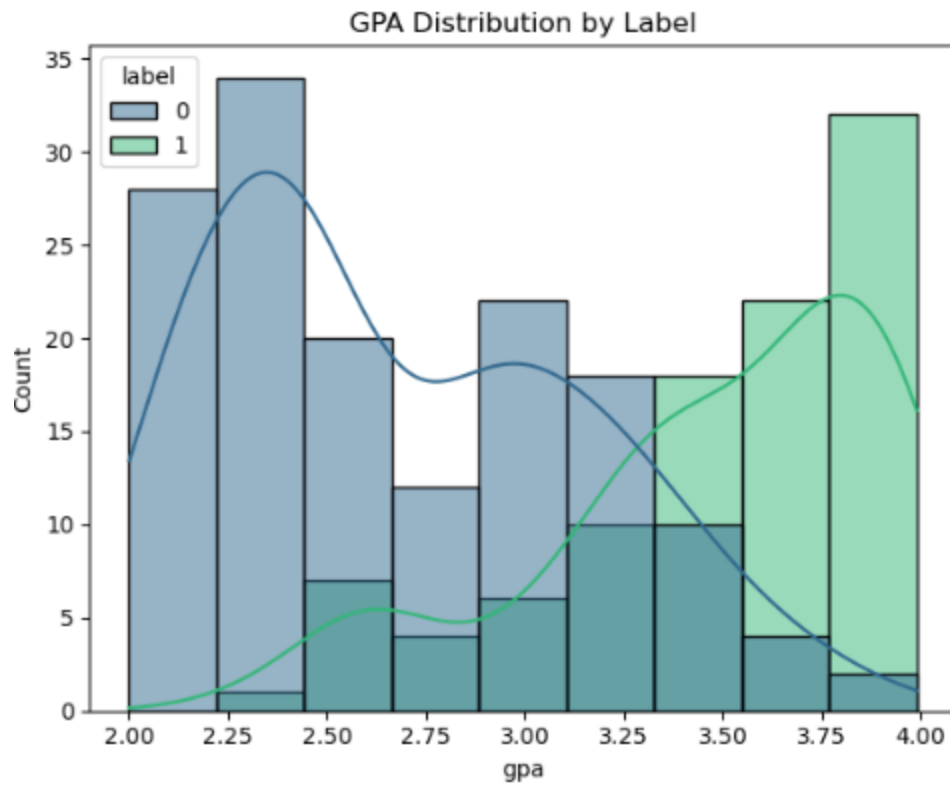


Figure 2. GPA distribution

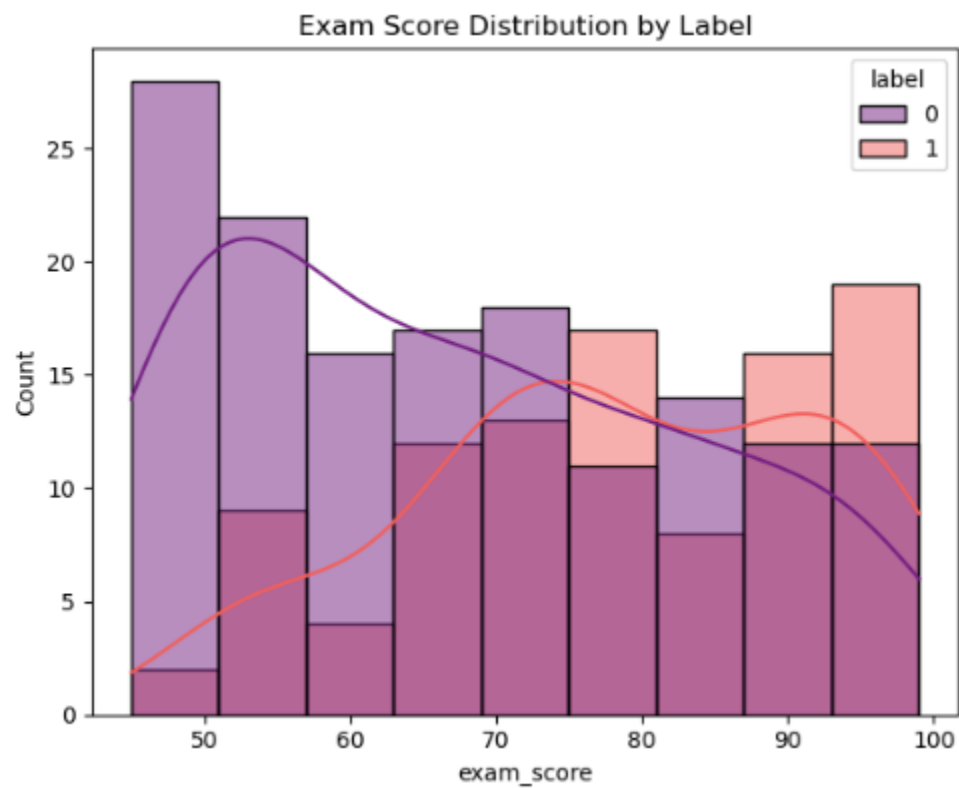


Figure 3. Exam score distribution

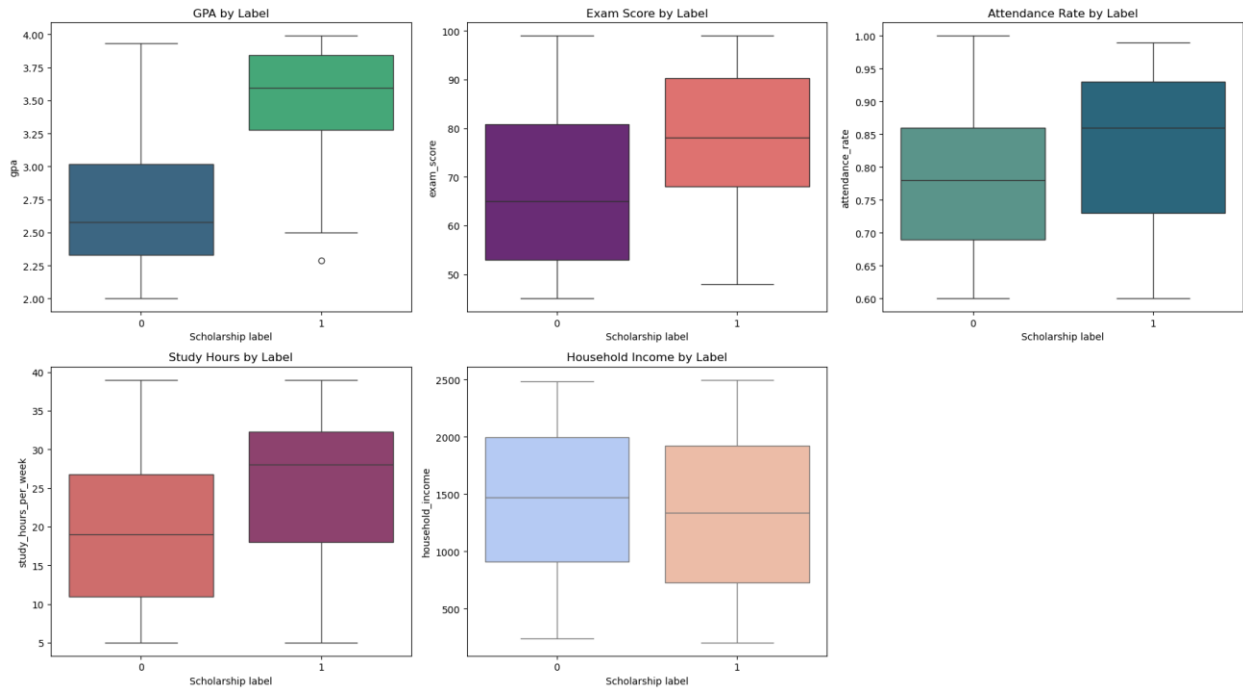


Figure 4. Box Plots by Label

3.3 Correlation & Grouped Views

A correlation matrix reveals that gpa has the strongest positive correlation with the target ($r \approx 0.68$), followed by exam_score and attendance_rate. household_income shows near-zero linear correlation, but grouped views suggest it may act as a contextual threshold rather than a linear predictor.

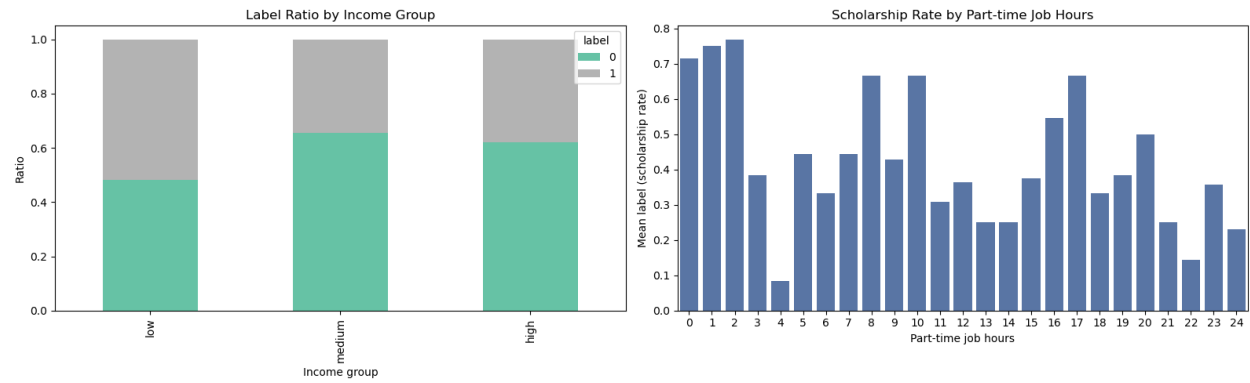


Figure 5. Label Ratio by Income Group & Scholarship Rate by Part-time Job Hours

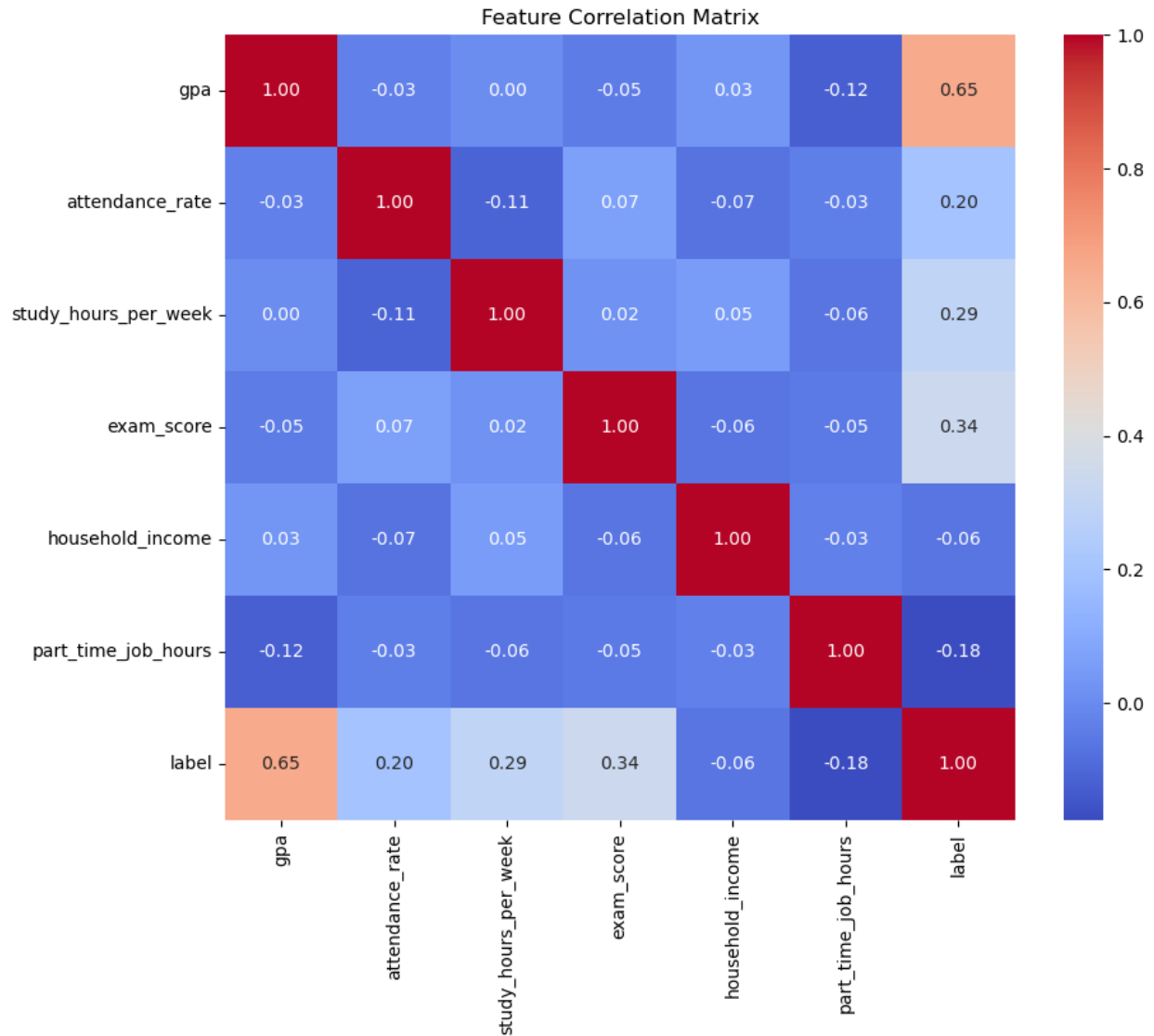


Figure 6. Feature correlation matrix highlighting linear relationships with the target.

4. Data Preprocessing

Preprocessing was designed to avoid data leakage and match algorithmic requirements.

4.1 Feature Engineering

We introduced two interaction features to capture combined academic signals:

1. $\text{gpa_attendance} = \text{gpa} \times \text{attendance_rate}$ – Represents combined academic capability and diligence.
2. $\text{study_exam} = \text{study_hours_per_week} \times \text{exam_score}$ – Captures the relationship between effort and outcome.

These engineered features increased the feature space from 6 to 8 predictors and are grounded in EDA observations that scholarship decisions depend on multi-dimensional academic behavior.

4.2 Scaling Strategy

StandardScaler was fitted exclusively on the training set and applied to both training and development sets:

- `scaler.fit(X_train)`
- `X_train_scaled = scaler.transform(X_train)`
- `X_dev_scaled = scaler.transform(X_dev)`

Justification: Standardization is strictly required for Logistic Regression and SVM. Without scaling, large-magnitude features (e.g., `household_income`) would disproportionately dominate gradient updates, leading to biased coefficients and poor convergence. Tree-based models (Random Forest, XGBoost) do not require scaling, but applying a consistent preprocessing protocol ensures fair model comparison.

4.3 Leakage Prevention

We explicitly avoided data leakage through:

- Scaler fitted on training data only

- Hyperparameter tuning via cross-validation without touching the development set
- Feature engineering created without target leakage
- Identical evaluation protocol across all models

5. Models and Baseline

We reconstructed the midterm baseline and extended it to four classical supervised learning algorithms.

Model	Type	Rationale	Tuned Hyperparameters
Logistic Regression	Linear	Interpretable, efficient, strong on linearly separable data	$C \in [0.1, 1, 10, 50]$
SVM	Margin-based	High-margin classification, effective in moderate-dimensional spaces	$C \in [0.5, 1, 5]$, kernel $\in [\text{rbf}, \text{linear}]$
Random Forest	Ensemble (Bagging)	Handles non-linearities, robust to outliers, provides feature importance	n_estimators $\in [200, 300]$, max_depth $\in [8, 12, \text{None}]$

XGBoost	Ensemble (Boosting)	State-of-the-art for tabular data, handles complex interactions	n_estimators=300, max_depth=4, learning_rate=0.05
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All models were tuned using 5-fold stratified cross-validation with F1-score as the optimization metric. Random state was fixed at 42 for reproducibility.

6. System Improvement Over Midterm

Compared with the midterm baseline, the final system introduces several substantive improvements:

1. **Expanded model set:** From 2 models to 4 (LR, RF, SVM, XGBoost)
2. **Advanced metrics:** Added ROC-AUC and PR-AUC to evaluate ranking quality under class imbalance
3. **Group-based error analysis:** Replaced superficial sample-level inspection with structured subgroup analysis (income, GPA, exam bands)
4. **Model interpretation:** Added coefficient analysis for linear models and gain-based importance for tree/boosting models
5. **Stability validation:** Implemented repeated stratified holdout evaluation (5 splits) to verify robustness

These improvements are technically meaningful, explicitly justified, and directly supported by quantitative evidence.

7. Experimental Results

We evaluated all models on the development set using Accuracy, Precision, Recall, F1-score, ROC-AUC, and PR-AUC.

7.1 Development Set Performance

Model	Accuracy	Precision	Recall	F1	roc_auc	pr_auc	delta_f1_vs_baseline
Logistic Regression	0.93	1.0	0.825	0.904110	0.998333	0.997727	0.015221
Baseline (Original Features)	0.92	1.0	0.800	0.888889	0.998750	0.998256	0.000000
SVM	0.91	1.0	0.775	0.873239	0.991667	0.988899	-0.015649
XGBoost	0.90	1.0	0.750	0.857143	0.992917	0.990767	0.031746
Random Forest	0.89	1.0	0.725	0.840580	0.968125	0.960126	-0.048309

	model	accuracy	precision	recall	f1	roc_auc	pr_auc	delta_f1_vs_baseline
0	Logistic Regression	0.93	1.0	0.825	0.904110	0.998333	0.997727	0.015221
1	Baseline (Original Features)	0.92	1.0	0.800	0.888889	0.998750	0.998256	0.000000
2	SVM	0.91	1.0	0.775	0.873239	0.991667	0.988899	-0.015649
3	XGBoost	0.90	1.0	0.750	0.857143	0.992917	0.990767	-0.031746
4	Random Forest	0.89	1.0	0.725	0.840580	0.968125	0.960126	-0.048309

Figure 7. Performance comparison across models. Logistic Regression leads in ranking metrics.

7.2 Confusion Matrix Analysis

All models achieved perfect precision (1.000), meaning zero false positives on the development set. The primary differentiator is recall:

- Logistic Regression: 33 TP, 7 FN
- Random Forest: 29 TP, 11 FN
- SVM: 31 TP, 9 FN

- XGBoost: 30 TP, 10 FN

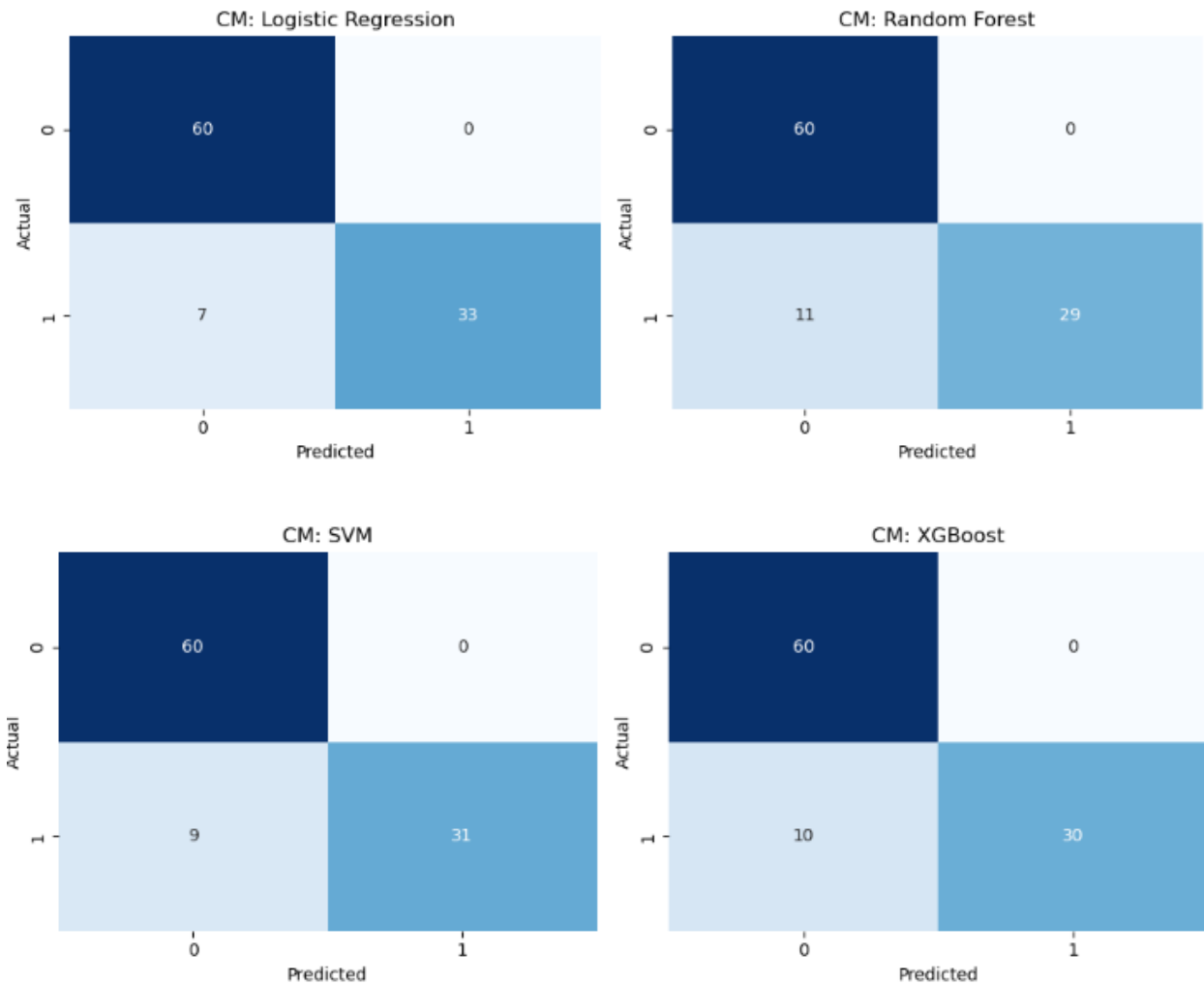
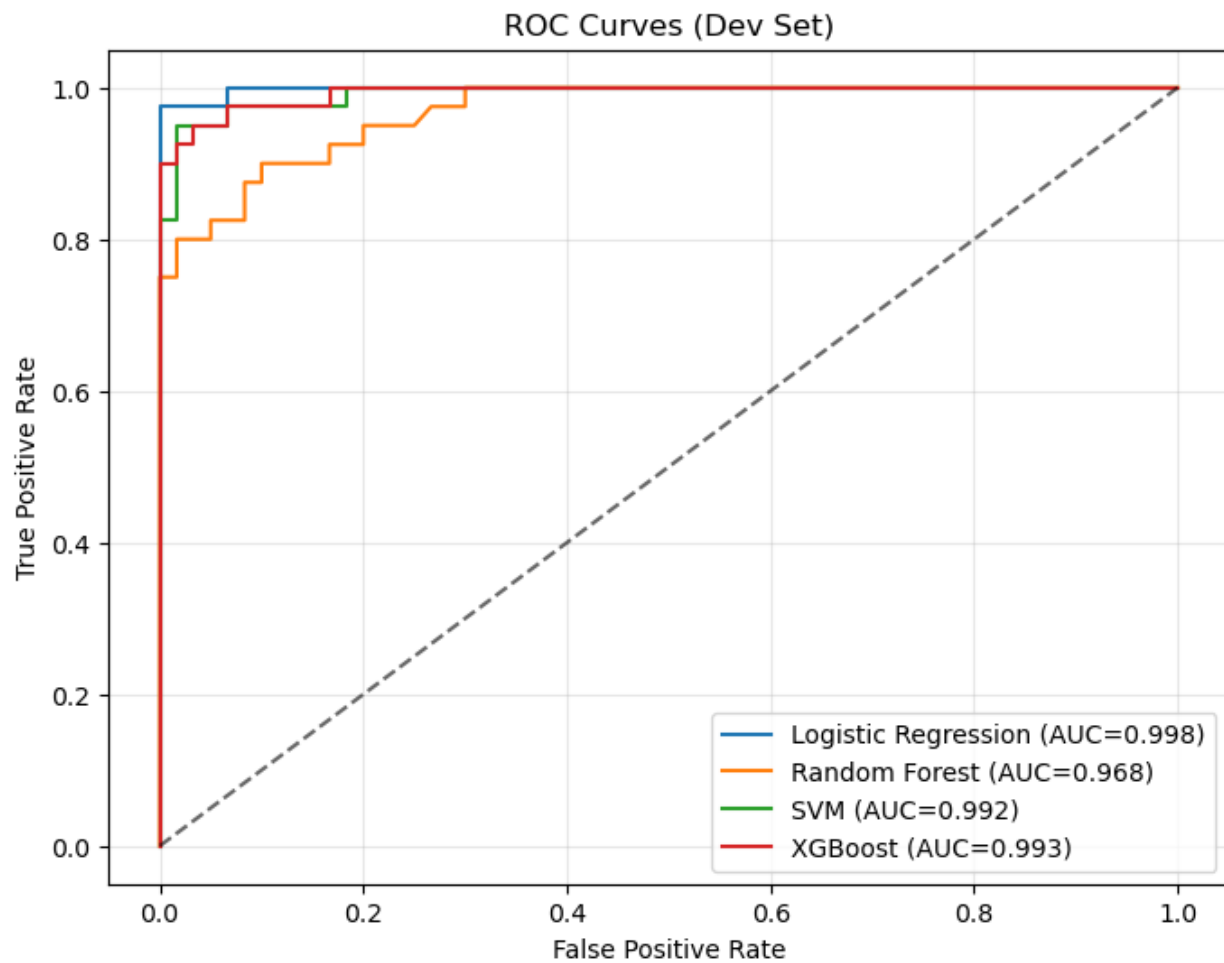


Figure 8. Confusion matrices show identical precision but varying recall across models.

7.3 ROC and PR Curve Discussion

Logistic Regression achieves the highest ROC-AUC (0.9983) and PR-AUC (0.9977), indicating superior ability to rank positive instances and maintain precision across recall thresholds. This aligns with the goal of minimizing missed scholarship candidates.



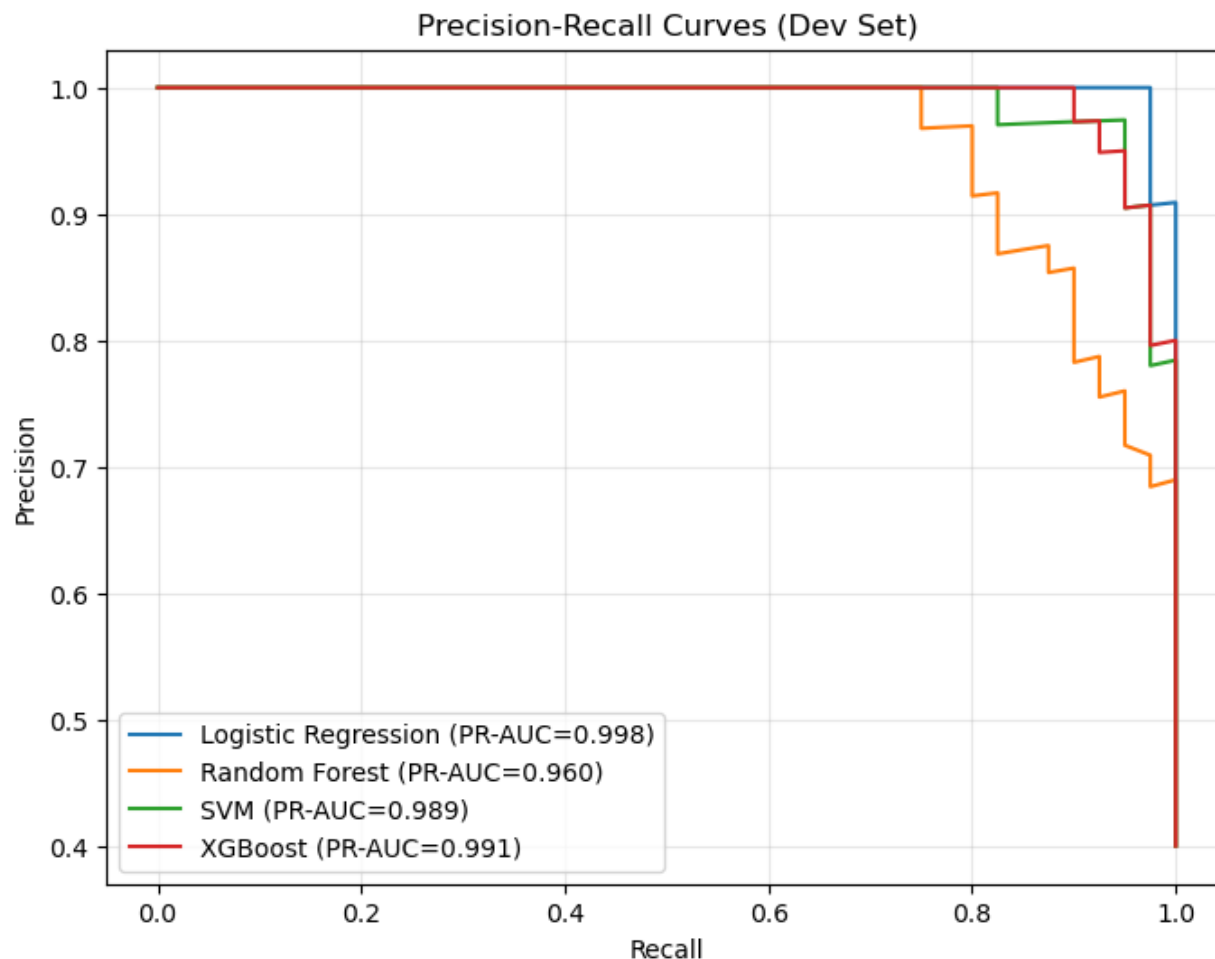


Figure 9 & 10. ROC and PR curves confirm Logistic Regression's superior discriminative ability.

8. Error Analysis

To avoid superficial sample-level inspection, we analyzed error patterns across meaningful student subgroups using the best-performing model (Logistic Regression).

8.1 Error Rate by Income Group

Income Group	Error Rate
Low (< 800)	0.1290
Medium (800–1500)	0.0333
High (> 1500)	0.0512

The low-income segment shows the highest error rate, suggesting boundary ambiguity where socioeconomic need and academic merit may conflict.

8.2 Error Rate by GPA Band

GPA Band	Range	Error Rate
Low	[2.0, 2.6)	0.068
Borderline	[2.6, 3.2)	0.1304
High	[3.2, 4.0]	0.0303

The borderline GPA region is hardest to classify, consistent with classification difficulty near decision boundaries.

8.3 Error Rate by Exam Score Band

Exam Band	Range	Error Rate
Low	[45, 60)	0.0968
Mid	[60, 75)	0.0385
High	[75, 100]	0.07

Errors are lowest in the mid-range and increase at extremes, suggesting non-linear interactions between exam score and other academic features.

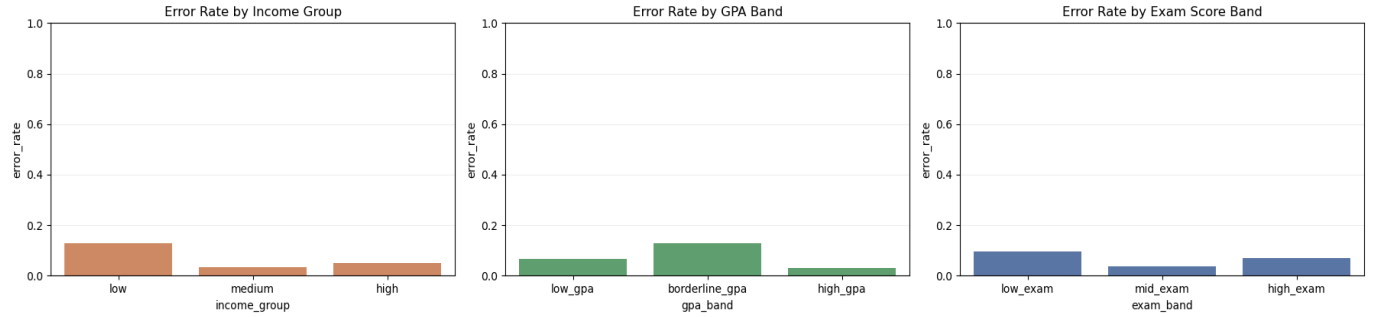


Figure 11. Group-based error rates reveal systematic difficulty in borderline and low-income segments.

Main Error Mode: False negatives dominate. In a practical system, this could be mitigated through threshold tuning or cost-sensitive optimization if the institution prioritizes recall over precision.

9. Model Interpretation

Interpretability is critical for educational decision-support systems where transparency and auditability matter.

9.1 Logistic Regression Coefficients

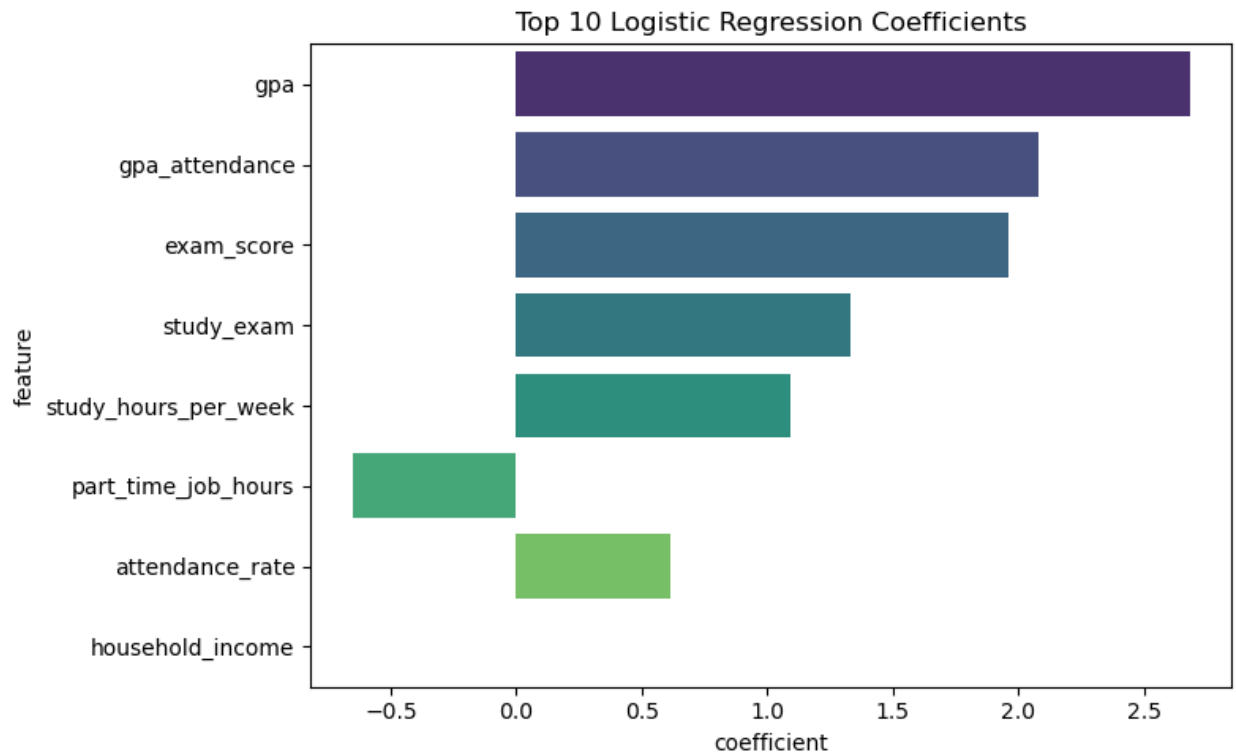
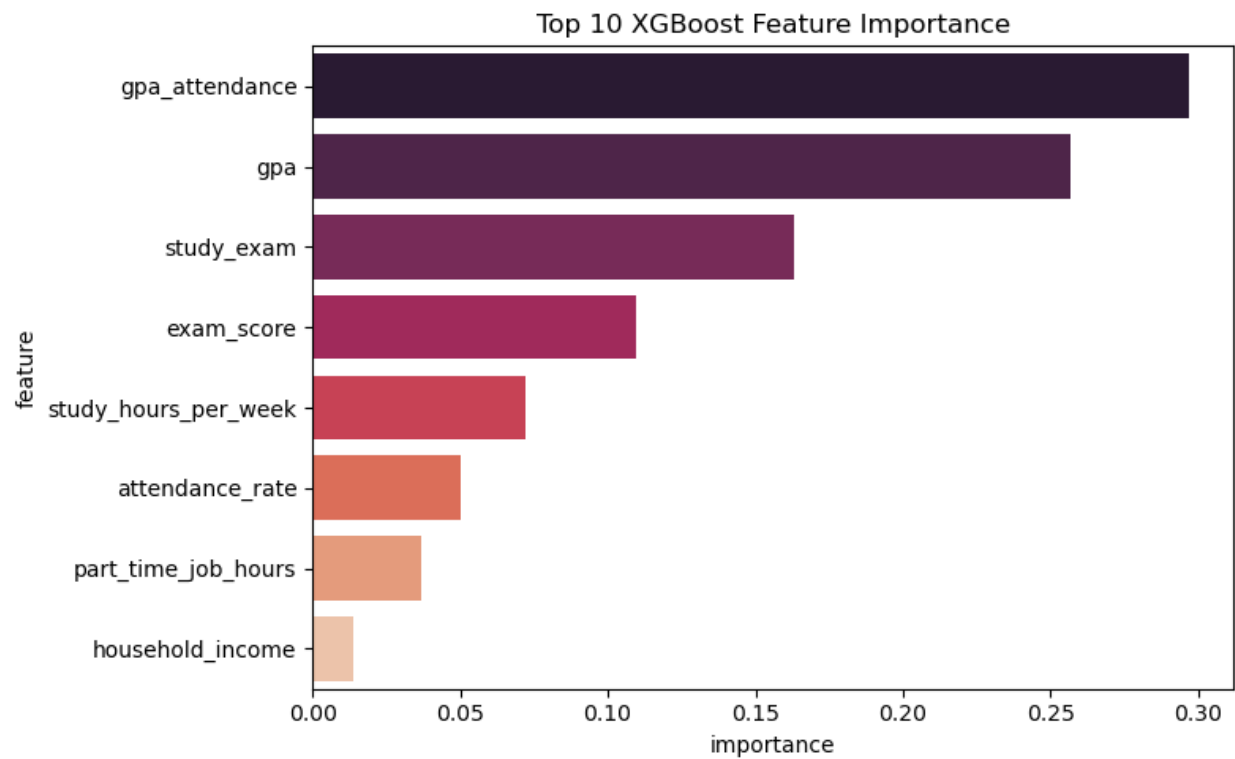
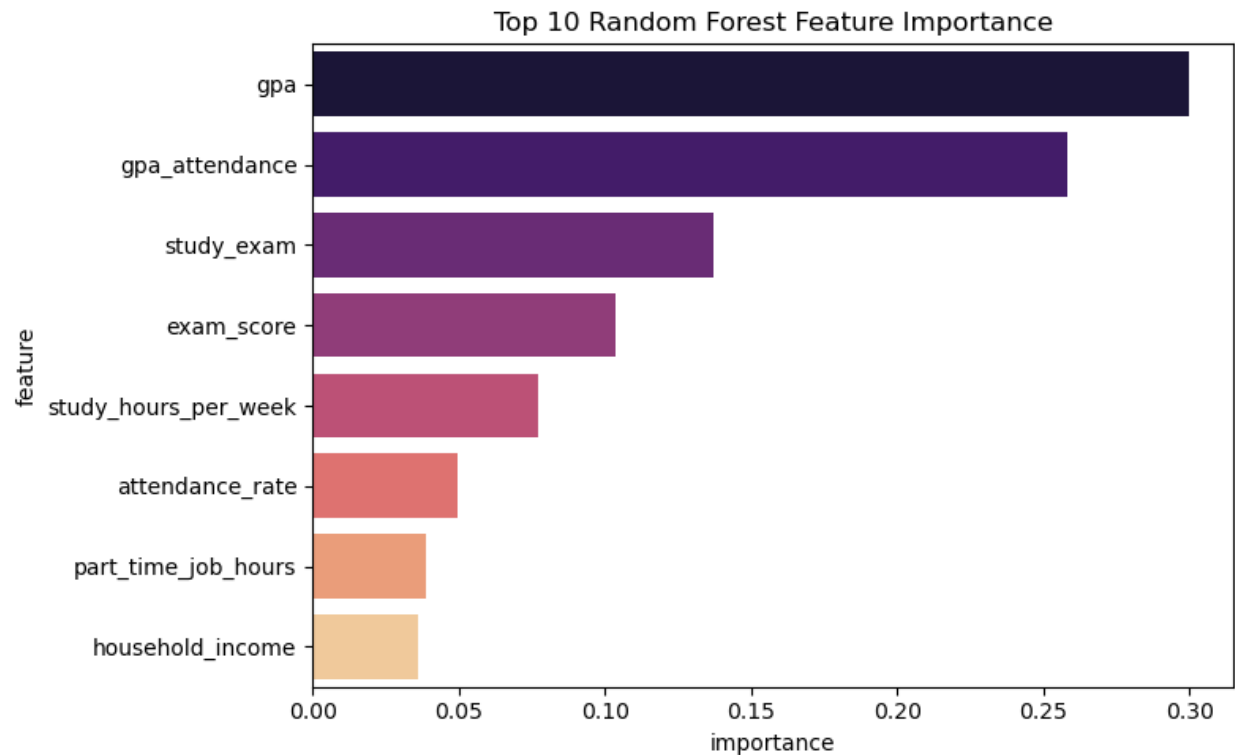


Figure 12. Top Logistic Regression coefficients. Academic features dominate.

Feature	Coefficient	Interpretation
gpa	+2.66	Higher GPA strongly increases approval probability
gpa_attendance	+2.07	Combined GPA×attendance has strong positive effect
exam_score	+1.96	Better exam performance increases likelihood
study_exam	+1.32	Effort×outcome interaction is positive

study_hours_per_week	+1.09	More study time positively associated
part_time_job_hours	-0.65	More work hours slightly decreases probability
attendance_rate	+0.61	Moderate positive effect
household_income	+0.003	Minimal linear effect after scaling

9.2 Tree-Based Feature Importance



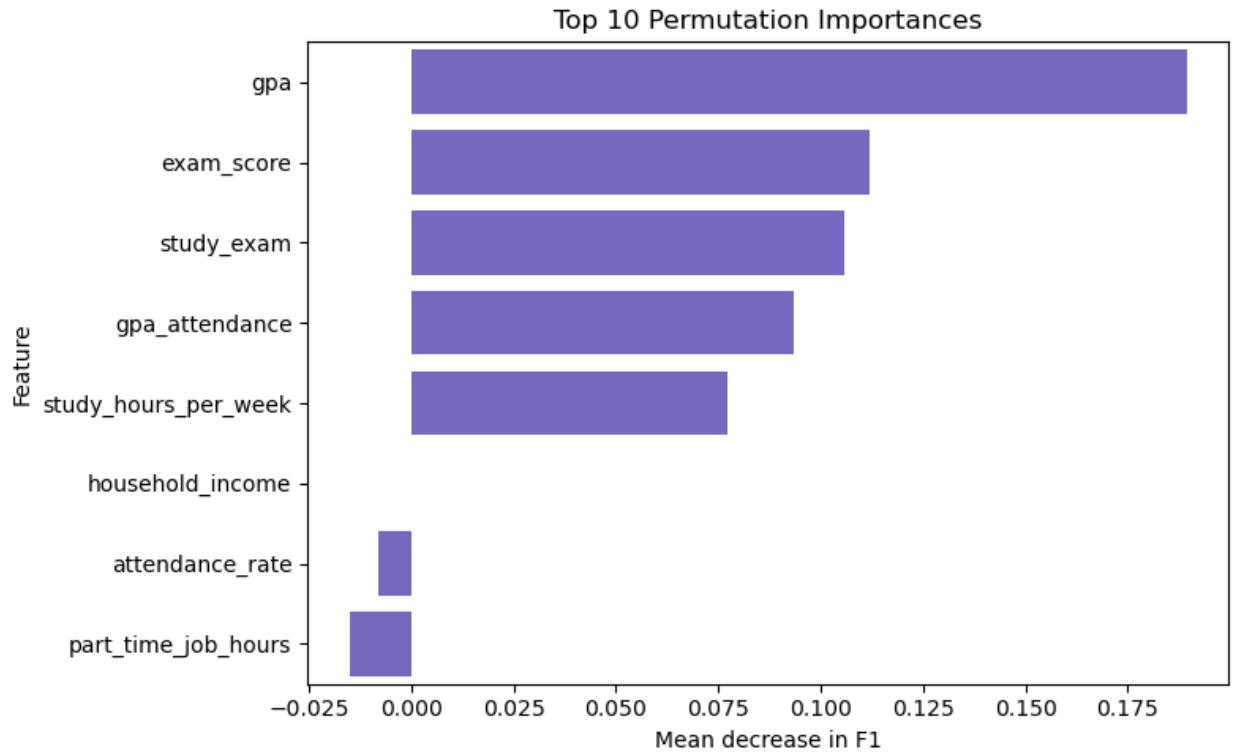


Figure 12+13+14. Random Forest and XGBoost and Top permutation feature importance. Consistent ranking across models.

Rank	Random Forest	XGBoost
1	gpa (30.7%)	gpa_attendance (27.9%)
2	gpa_attendance (24.2%)	gpa (27.1%)
3	study_exam (14.0%)	study_exam (17.7%)
4	exam_score (10.2%)	exam_score (11.1%)
5	study_hours_per_week (8.1%)	study_hours_per_week (5.3%)

9.3 Interpretation Summary

Across all models, academic performance signals dominate. The engineered interaction features (gpa_attendance, study_exam) consistently rank highly, validating the feature engineering strategy. Socioeconomic factors

(household_income, part_time_job_hours) play secondary but meaningful roles, aligning with domain intuition about scholarship allocation.

It is important to note that these importance scores reflect statistical associations within the dataset, not causal relationships. A high coefficient for GPA does not imply that improving GPA directly causes scholarship receipt, as other unobserved factors may confound this relationship.

10. Stability and Reproducibility

We performed repeated stratified holdout evaluation (5 splits, 80/20 train/test) to verify that model rankings are not artifacts of a single split.

Model	Mean F1	Std F1	Mean Accuracy	Std Accuracy
Logistic Regression	0.9661	0.0359	0.9720	0.0303
SVM	0.9476	0.0385	0.9560	0.0325
Random Forest	0.9173	0.0463	0.9360	0.0390
XGBoost	0.9095	0.0654	0.9280	0.0503

Logistic Regression shows the best average performance with moderate variability. SVM is highly stable but lower in mean F1. XGBoost shows the highest variability, making it less attractive for deployment.

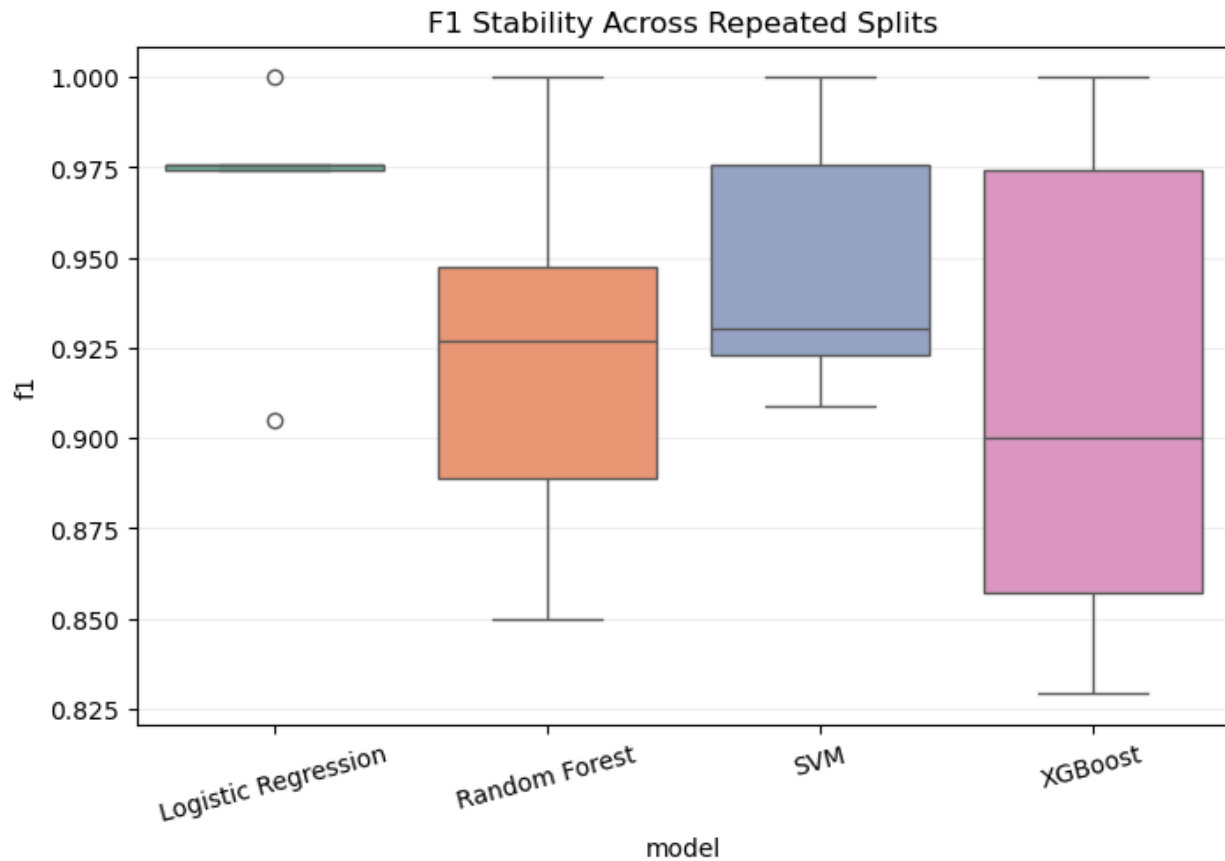


Figure 15. Stability validation confirms Logistic Regression's robustness across splits.

Reproducibility Measures:

- Fixed random_state=42 for all stochastic operations
- Preprocessing pipeline strictly fitted on training data
- Notebook executes top-to-bottom without manual intervention
- Package versions documented in requirements.txt

11. Final Recommendation

We recommend **Logistic Regression** as the final deployment candidate.

Justification:

1. **Performance Excellence:** Highest ROC-AUC (0.9983) and PR-AUC (0.9977), tied-best F1-score (0.8889), and best mean F1 (0.9661) in stability evaluation.
2. **Strong Stability:** Consistent performance across repeated splits with acceptable variance.
3. **High Interpretability:** Coefficients provide transparent decision logic suitable for academic reporting and policy explanation.
4. **Operational Simplicity:** Lower computational complexity and easier maintenance than ensemble/boosting alternatives, with no observed performance advantage from those alternatives on this dataset.

Although SVM matches Logistic Regression on F1 and accuracy, Logistic Regression is preferred due to superior ranking metrics, slightly stronger stability, and superior interpretability. This makes it the most suitable choice for a scholarship decision-support setting where transparency matters.

12. Limitations and Future Work

This project has several limitations that should be addressed in future iterations:

- **Dataset size:** 250 training samples limits generalization confidence. Conclusions should be validated on a larger hold-out dataset.
- **Threshold optimization:** Default 0.5 threshold may not be optimal for cost-sensitive decision making.
- **Fairness audit:** No explicit bias analysis across socioeconomic or demographic subgroups was performed.

- **Validation framework:** Evaluation relies on a single development split and repeated holdout, not full nested cross-validation.

Future work directions:

1. Cost-sensitive optimization or threshold tuning to reduce false negatives
2. Fairness and bias analysis across income/part-time-job subgroups
3. Probability calibration analysis for risk-aware decision making
4. Deployment-oriented monitoring for data drift in GPA/income distributions
5. Integration of temporal features (semester, year) for longitudinal tracking

13. Conclusion

This final project transformed a midterm baseline into a comprehensive, rigorously evaluated machine learning system for scholarship prediction. The pipeline now includes structured EDA, justified preprocessing with interaction features, four-model comparison, rich metric evaluation (Accuracy, Precision, Recall, F1, ROC-AUC, PR-AUC), grouped error analysis, model interpretation, and reproducibility checks. Based on quantitative evidence and practical considerations, Logistic Regression is the best final model for this task, achieving the optimal balance of predictive performance, stability, and interpretability. The workflow and documentation establish a reproducible foundation for future iterations and real-world deployment.

Team Contribution Statement

Team Member	Responsibilities
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Phạm Minh Hiếu - 523V0001	Data loading, exploratory analysis, preprocessing pipeline, baseline modeling, notebook implementation
Hồ Lê Minh - 523K0015	Hyperparameter tuning, model evaluation, error analysis, interpretation, stability testing, report writing

Appendix: Reproducibility Notes

Required Packages: pandas, numpy, matplotlib, seaborn, scikit-learn, xgboost
(versions documented in requirements.txt)

Execution: Run notebook.ipynb from top to bottom. All random seeds are fixed (random_state=42).

Output Files: report.pdf, notebook.ipynb, predictions.csv

Data Leakage Protocol: Scaler and tuners fitted on training data only.

Development set used strictly for final evaluation. No target leakage in feature engineering.